

PARITY REFUTES THE PRINTED FOUR-TERM AMICABLE-TOTIENT EQUATIONS

ABSTRACT. For an amicable pair (A, B) with $A = 2^n \prod_i p_i$ and $B = 2^n \prod_j q_j$, Mavaddat and Alikhani put $S = \varphi(A) + \varphi(B)$, $A - S = 2^n x$, $B - S = 2^n y$, and conjecture for $k = m = 4$, $A = 2^n abcd$, $B = 2^n efgh$, that $x - 1 = [ab(c + d) + cd] - (e + f + g + h)$ and $y - 1 = [ef(g + h) + gh] - (a + b + c + d)$. We show that these two equations, as printed, fail for every amicable pair of type $(4, 4)$ with $\gcd(A, B) = 2^n$ exactly, and for every labelling of the eight primes: x and y are odd, while each printed right-hand side is odd, so each equation demands that x or y be even. Neither right-hand side is a symmetric function of its letters, and we do not attempt to settle whatever symmetric statement may have been intended. On the amicable pair $A = 32642324$, $B = 35095276$ we also exclude the sixteen typo-charitable variants of the printed equations obtained by exchanging their brackets, their subtracted sums or their two sides, or by deleting the offsets -1 — parity alone does not reach the offset-free readings. We record the four-term identity $x = e_3(q) + e_1(q) - e_2(p) - 1$, an elementary consequence of $\prod_i (p_i + 1) = \prod_j (q_j + 1)$ and not a new theorem.

1. THE PRINTED EQUATIONS

Let (A, B) be an amicable pair, so $\sigma(A) = \sigma(B) = A + B$ with $A \neq B$. In [4], A. R. Mavaddat and S. Alikhani study such pairs through the Euler totient. Writing

$$(1) \quad S = \varphi(A) + \varphi(B), \quad A - S = 2^n x, \quad B - S = 2^n y,$$

they prove identities for x and y in the cases $A = 2^n ab$, $B = 2^n cd$ (their Theorem 2.1), $A = 2^n abc$, $B = 2^n de$ (Theorem 2.3), and $A = 2^n abc$, $B = 2^n def$ (Theorem 2.5). The quantities x and y are theirs; nothing outside [4] defines them. In the concluding section they propose “the following general rules for an amicable pair (A, B) where $A = 2^n \prod_{i=1}^k p_i$ and $B = 2^n \prod_{j=1}^m q_j$ as a conjecture”. The second of the two conjecture environments there is the case $k = m = 4$:

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Conjecture ([4], Sec. 3; *Case: Symmetric 4-term*, $A = 2^n abcd$, $B = 2^n efgh$). *For symmetric higher-order pairs, the relationship shifts to a comparison of their symmetric polynomials:*

$$(2) \quad y - 1 = [ef(g + h) + gh] - (a + b + c + d),$$

$$(3) \quad x - 1 = [ab(c + d) + cd] - (e + f + g + h).$$

We read $a, \dots, h$ as odd primes, with $abcd$ and $efgh$ the odd parts of A and B , each a product of four distinct odd primes, and $n \geq 1$. In the classification of Costello [2] and Garcia, Pedersen and te Riele [3], both cited in [4], such a pair has type (4, 4).

Despite the heading word *Symmetric*, neither right-hand side is a symmetric function of its letters: $ab(c + d) + cd = abc + abd + cd$ mixes two of the four monomials of e_3 with one of the six of e_2 . What follows concerns (2)–(3) exactly as printed. We neither reconstruct nor settle any symmetric statement that may have been intended, and a reader who takes the display for a misprint should take this note as settling the printed formulas only. No order on the primes is imposed anywhere in [4], so the display admits a strict reading (*every* labelling of the eight primes) and a charitable one (*some* labelling, $4! \times 4! = 576$ per pair). Everything below refutes the charitable reading, hence both.

2. THE PARITY OF x AND y

Lemma 1. *Let $n \geq 1$ and let $A = 2^n M$, $B = 2^n N$ with M, N odd, each of M and N having at least two distinct prime factors. Define S , x , y by (1). Then $2^n \mid A - S$ and $2^n \mid B - S$, and both x and y are odd. Neither amicability nor $\gcd(A, B) = 2^n$ is assumed.*

Proof. Since $\gcd(2^n, M) = 1$ and $\varphi(2^n) = 2^{n-1}$, we have $\varphi(A) = 2^{n-1}\varphi(M)$ and $\varphi(B) = 2^{n-1}\varphi(N)$, so $S = 2^{n-1}(\varphi(M) + \varphi(N))$ and

$$(4) \quad A - S = 2^n \left(M - \frac{\varphi(M) + \varphi(N)}{2} \right), \quad x = M - \frac{\varphi(M) + \varphi(N)}{2},$$

and symmetrically $y = N - (\varphi(M) + \varphi(N))/2$. If M has $r \geq 2$ distinct odd prime factors then $2^r \mid \varphi(M)$, since φ is multiplicative and $p - 1$ is even for each odd prime p ; so $4 \mid \varphi(M)$ and $4 \mid \varphi(N)$, the quotient $(\varphi(M) + \varphi(N))/2$ is an even integer, and $x \equiv M \equiv 1 \pmod{2}$. The same computation gives y odd. $\square$

All the proof uses is $n \geq 1$ together with $4 \mid \varphi(M)$ and $4 \mid \varphi(N)$. The hypothesis on the number of prime factors cannot be dropped: $M = 3$, $N = 5$ give $(\varphi(M) + \varphi(N))/2 = 3$ and hence $x = 0$, even.

Theorem 2. *Let (A, B) be any amicable pair of type (4, 4) with $\gcd(A, B) = 2^n$ exactly, $n \geq 1$; that is, $A = 2^n abcd$ and $B = 2^n efgh$ with $a, \dots, h$ odd primes, $\{a, b, c, d\}$ and $\{e, f, g, h\}$ each of size four. Then neither (2) nor (3) holds, for any labelling of the eight primes. The same is true after*

transposing x and y . In particular the two printed equations fail for every such pair, for every $n \geq 1$, with no bound on A .

Proof. By Lemma 1, x and y are odd, so $x - 1$ and $y - 1$ are even. On the right of (3), $ab(c + d) + cd = abc + abd + cd$ is a sum of three products of odd numbers, hence odd, while $e + f + g + h$ is a sum of four odd numbers, hence even; so the right-hand side is odd $-$ even $=$ odd $\neq x - 1$. The right-hand side of (2) has the same shape, $ef(g + h) + gh = efg + efh + gh$ odd and $a + b + c + d$ even, so it too is odd and cannot equal $y - 1$. Both brackets are odd and both subtracted sums are even, so all four ways of pairing $x - 1$ or $y - 1$ against either right-hand side fail identically; in particular transposing x and y changes nothing. No step refers to the labelling, so all 576 labellings fail at once. $\square$

Remark. Theorem 2 contradicts nothing proved in [4]. Its Theorem 2.1 gives $x = c + d - 1$, odd; its Theorem 2.3 gives $x = (d + e) - (a + b + c)$, even $-$ odd, odd; its Theorem 2.5 gives $x - 1 = [de + df + ef] - (a + b + c)$, odd $-$ odd, even, so again x is odd. All three are consistent with Lemma 1.

3. A WITNESS FOR THE OFFSET-FREE READINGS

The offsets -1 in (2)–(3) are not used consistently in [4]: the paper's first conjecture environment, the type-(4, 2) case, and its Theorem 2.3 carry no offset, while Theorem 2.5 and the display above do. A reader may therefore take the display without the offsets, and parity does not reach that reading: dropping the -1 makes the required bracket value odd, which is admissible. Rather than guess which typo was made, we settle every variant of the printed equations at once. Put

$$\begin{aligned} \mathcal{B}_A &= ab(c + d) + cd, & \mathcal{B}_B &= ef(g + h) + gh, \\ \Sigma_A &= a + b + c + d, & \Sigma_B &= e + f + g + h, \end{aligned}$$

and call an equation *admissible* if it has the form

$$(5) \quad L = \mathcal{B} - \Sigma, \quad L \in \{x - 1, y - 1, x, y\}, \quad \mathcal{B} \in \{\mathcal{B}_A, \mathcal{B}_B\}, \quad \Sigma \in \{\Sigma_A, \Sigma_B\}.$$

There are sixteen admissible equations. The two printed ones (2) and (3) are among them; so is each variant obtained by transposing x and y , by swapping the two brackets, by swapping the two subtracted sums, by deleting the offsets, or by any combination.

Theorem 3. *Let*

$$A = 2^2 \cdot 11 \cdot 13 \cdot 149 \cdot 383 = 32642324, \quad B = 2^2 \cdot 17 \cdot 47 \cdot 79 \cdot 139 = 35095276.$$

Then (A, B) is an amicable pair of type (4, 4) with $\gcd(A, B) = 4$ exactly, $x = 807269$ and $y = 1420507$, and no one of the sixteen admissible equations (5) holds for any labelling with $\{a, b, c, d\} = \{11, 13, 149, 383\}$ and $\{e, f, g, h\} = \{17, 47, 79, 139\}$.

Proof. The factorisations are checked by multiplication: $11 \cdot 13 = 143$, $149 \cdot 383 = 57067$, $143 \cdot 57067 = 8160581$, $4 \cdot 8160581 = 32642324$; and $17 \cdot 47 = 799$, $79 \cdot 139 = 10981$, $799 \cdot 10981 = 8773819$, $4 \cdot 8773819 = 35095276$. All eight odd primes are distinct, so both odd parts are squarefree with four prime factors and $\gcd(A, B) = 2^2$ exactly. Amicability:

$$\begin{aligned}\sigma(A) &= 7 \cdot 12 \cdot 14 \cdot 150 \cdot 384 = 67737600, \\ \sigma(B) &= 7 \cdot 18 \cdot 48 \cdot 80 \cdot 140 = 67737600, \\ A + B &= 32642324 + 35095276 = 67737600.\end{aligned}$$

Next $\varphi(A) = 2 \cdot 10 \cdot 12 \cdot 148 \cdot 382 = 13568640$ and $\varphi(B) = 2 \cdot 16 \cdot 46 \cdot 78 \cdot 138 = 15844608$, so $S = 29413248$, $A - S = 3229076 = 4 \cdot 807269$ and $B - S = 5682028 = 4 \cdot 1420507$, giving $x = 807269$ and $y = 1420507$; both are odd, as Lemma 1 requires.

The bracket $[pq(r + s) + rs]$ is unchanged by swapping $p \leftrightarrow q$ or $r \leftrightarrow s$, so over a four-element set it takes at most $\binom{4}{2} = 6$ values, each realised by exactly 4 of the 24 orderings. Over $\{11, 13, 149, 383\}$ the six values are

$$133143, \quad 654023, \quad 684443, \quad 767391, \quad 798279, \quad 1369751,$$

computed for instance as $143 \cdot 532 + 57067 = 133143$ and $57067 \cdot 24 + 143 = 1369751$; over $\{17, 47, 79, 139\}$ they are

$$185163, \quad 256331, \quad 301451, \quad 581591, \quad 628511, \quad 703583,$$

the largest being $10981 \cdot 64 + 799 = 703583$. All twelve values are odd. Finally $\Sigma_A = 556$ and $\Sigma_B = 282$, both even.

An admissible equation $L = \mathcal{B} - \Sigma$ asserts $\mathcal{B} = L + \Sigma$, so it is enough to compare the eight numbers $L + \Sigma$ with the two six-element sets above. For $L \in \{x - 1, y - 1\} = \{807268, 1420506\}$ the value $L + \Sigma$ is even, while every achievable $\mathcal{B}$ is odd: this disposes of eight equations at once, in agreement with Theorem 2. For $L \in \{x, y\}$ the value $L + \Sigma$ is odd, so parity is silent and we compare values:

$$\begin{aligned}x + \Sigma_B &= 807551, & x + \Sigma_A &= 807825, \\ y + \Sigma_B &= 1420789, & y + \Sigma_A &= 1421063.\end{aligned}$$

None of the four is one of the six values over $\{11, 13, 149, 383\}$, and all four exceed 703583, the largest value over $\{17, 47, 79, 139\}$. That is the remaining eight equations. $\square$

4. THE CORRECTED IDENTITY

Proposition 4. *Let (A, B) be an amicable pair with $A = 2^n M$, $B = 2^n N$, $n \geq 1$, $M = p_1 p_2 p_3 p_4$ and $N = q_1 q_2 q_3 q_4$ squarefree and odd. Write $E_k = e_k(p)$ and $F_k = e_k(q)$ for the elementary symmetric functions. Then*

$$(6) \quad x = F_3 + F_1 - E_2 - 1, \quad y = E_3 + E_1 - F_2 - 1.$$

Proof. From $\sigma(A) = \sigma(B)$ and $\sigma(2^n) = 2^{n+1} - 1$ we get $\prod_i(p_i + 1) = \prod_j(q_j + 1)$; call this common value T , so

$$T = E_4 + E_3 + E_2 + E_1 + 1 = F_4 + F_3 + F_2 + F_1 + 1.$$

Expanding $\varphi(M) = \prod_i(p_i - 1)$ gives $\varphi(M) = E_4 - E_3 + E_2 - E_1 + 1 = T - 2(E_3 + E_1)$, and likewise $\varphi(N) = T - 2(F_3 + F_1)$. Substituting into (4),

$$x = M - \frac{\varphi(M) + \varphi(N)}{2} = E_4 - T + (E_3 + E_1) + (F_3 + F_1) = F_3 + F_1 - E_2 - 1,$$

the last step by $E_4 - T = -(E_3 + E_2 + E_1 + 1)$. The formula for y is symmetric. $\square$

On the witness, $E_1 = 556$, $E_2 = 69978$, $E_3 = 1445684$, $F_1 = 282$, $F_2 = 25732$, $F_3 = 876966$, and (6) reads $876966 + 282 - 69978 - 1 = 807269 = x$ and $1445684 + 556 - 25732 - 1 = 1420507 = y$.

The only non-arithmetic input to Proposition 4 is $\prod_i(p_i + 1) = \prod_j(q_j + 1)$, which [4] itself derives twice and which is standard breeder machinery in the sense of Borho and Hoffmann [1]. We record (6) because it locates the error, not because it is difficult, and we claim it as a correction and not as a result: we did not obtain the full texts of [1], [2] or [3], and whether (6) is already in print we leave open. The bracket $[ab(c + d) + cd]$ is a factored writing of e_2 for *three* letters — three monomials, hence odd — carried typographically to *four* letters, where by (6) the coefficient is e_3 with four monomials, hence even; in the corrected form $x - 1 = F_3 + F_1 - E_2 - 2$ is even, as Lemma 1 demands.

5. SCOPE AND VERIFICATION

Theorem 2 settles the two printed equations, in both the strict and the charitable reading of the labelling, for every amicable pair of type (4, 4) with $\gcd(A, B) = 2^n$ exactly; Theorem 3 settles all sixteen admissible variants (5), including the offset-free ones, on one explicit pair. The offset-free variants in general are left open: parity does not apply to them and no general argument is offered here. The other conjecture environment of [4], the type-(4, 2) case, which carries no offsets, is not treated here, and no statement of [4] other than (2)–(3) is refuted here.

The program accompanying this note reads the witness exactly as printed above, in integer arithmetic only, and re-derives every number in Theorems 2–3 and Proposition 4, including all 576 labellings against all sixteen admissible equations (5), the parity lemma over an independent family of 4845 squarefree odd M that involves no amicable pair at all, seven further amicable pairs of type (4, 4) from their prime factors alone, and the three identities [4] proves. The program restates its own limits in a scope block; in particular it checks mathematics and not bibliography, so the wording quoted in Section 1 and the description of [4] are not re-verified there.

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